Slip 1

1. **Why is proper indentation required in Python?** It defines code blocks (like `if` statements and loops) instead of using curly braces.
2. **What exception occurs if Python code has mixed or inconsistent indentation?** An `IndentationError` is raised.
3. **Which graph search algorithm is used to solve the Water Jug problem?** Breadth-First Search (BFS).
4. **Why is a `visited` set maintained in state-space search problems?** To track explored states and prevent infinite loops.
5. **How does a Model-Based Agent differ from a Simple Reflex Agent?** It maintains internal memory/state to track parts of the environment not visible in the current perception.

Slip 2

1. **What is the base condition in the recursive Tower of Hanoi program?** When n = 1, move the single disk directly from the source to the destination peg.
2. **What is the time complexity of the Tower of Hanoi algorithm?** Exponential time complexity of O(2^n).
3. **In what type of applications is the Mini-Max algorithm used?** Two-player, zero-sum, perfect-information games like Tic-Tac-Toe and Chess.
4. **What are the roles of the Maximizing and Minimizing players in Mini-Max?** The Maximizer tries to maximize the score, while the Minimizer tries to minimize it.
5. **What do X and Y represent in Linear Regression?** X represents the independent feature variable and Y represents the dependent target variable.

Slip 3

1. **How does a Simple Reflex Agent make decisions?** Purely based on current perceptions using condition-action (IF-THEN) rules without history.
2. **What is the primary drawback of a Simple Reflex Agent?** It fails in partially observable environments because it lacks memory.
3. **What type of machine learning algorithm is Random Forest?** An ensemble learning method combining multiple decision trees.
4. **What does the `n_estimators=100` parameter specify in Random Forest?** The total number of decision trees built in the forest.
5. **What is Backtracking in the N-Queens problem?** A search strategy that abandons paths immediately when a constraint violation occurs.

Slip 4

1. **Why is Logistic Regression used instead of Linear Regression for classification?** It outputs discrete categorical class probabilities between 0 and 1 using a sigmoid function.
2. **What function maps linear outputs to probabilities in Logistic Regression?** The Sigmoid function.
3. **What is the primary purpose of Alpha-Beta Pruning?** To optimize Mini-Max game trees by eliminating branches that cannot influence the result.
4. **What do \alpha and \beta represent in Alpha-Beta pruning?** \alpha tracks the maximizer's best option, and \beta tracks the minimizer's best option.
5. **Which Python library is commonly used to create scatter plots?** Matplotlib (`matplotlib.pyplot`).

Slip 5

1. **What problem-solving paradigm does the Monkey Banana problem represent?** State-space search and automated goal-based planning.
2. **What sequence of actions does the monkey perform to reach the goal?** Move to box \rightarrow Push box under banana \rightarrow Climb box \rightarrow Grab banana.
3. **How must feature inputs be formatted when calling `.predict()` in Scikit-Learn?** As a 2D array or DataFrame.
4. **How does a rule-based chatbot function?** It matches user inputs against predefined pattern rules to provide fixed responses.
5. **Why is `.lower()` applied to chatbot user input?** To make input text comparisons case-insensitive.

Slip 6

1. **What type of algorithm is Hill Climbing?** A heuristic local search algorithm that moves toward increasing value to find a local optimum.
2. **What are the major limitations of the Hill Climbing algorithm?** Local maxima, plateaus, and ridges.
3. **How are built-in datasets accessed in Scikit-Learn?** Through the `sklearn.datasets` module.
4. **What does `plt.show()` do in Matplotlib?** Renders and opens the graphical plot window on screen.
5. **Why is a depth limit required when evaluating large game trees?** To bound search time and prevent infinite execution depth.

Slip 7

1. **Which bitwise operators combine multiple boolean filters in Pandas?** `&` (AND) and `|` (OR) enclosed within parentheses.
2. **What function checks if a string contains a specific substring in Pandas?** `df['col'].str.contains('pattern')`.
3. **Why does a Smart Home Security agent track previous door states?** To detect state transitions (e.g., changing from CLOSED to OPEN) rather than static states alone.
4. **Which data structure powers Breadth-First Search (BFS)?** A Queue operating on First-In, First-Out (FIFO) ordering.
5. **What is the time complexity of BFS on a graph with vertices V and edges E?** O(V + E).

Slip 8

1. **What is the initial location setup for the monkey, box, and banana?** Monkey at door, box at window, and banana at ceiling.
2. **What table summarizes classification prediction errors?** A Confusion Matrix.
3. **What metrics are used to evaluate classification performance?** Accuracy, Precision, Recall, and F1-score.
4. **How do you apply conditional updates to DataFrame rows without loops in Pandas?** Using vectorized conditional assignment with `.loc[]`.
5. **How do you display the first few rows of a DataFrame?** Using `df.head()`.

Slip 9

1. **Which Pandas method sorts a DataFrame by specific column values?** `df.sort_values(by='column_name')`.
2. **Does `sort_values()` modify the original DataFrame by default?** No, it returns a sorted copy unless `inplace=True` is specified.
3. **What is the Blocks World domain used for in AI?** Testing planning and heuristic state-space representation algorithms.
4. **What is the structural difference between BFS and DFS traversal?** BFS uses a Queue for level-order search, while DFS uses a Stack for deep-branch search.
5. **Which graph search algorithm guarantees finding the shortest path in unweighted graphs?** Breadth-First Search (BFS).

Slip 11

1. **How do you load a CSV file into a Pandas DataFrame?** Using `pd.read_csv('filename.csv')`.

2. **What attribute stores the slope coefficients in Scikit-Learn Linear Regression?** `model.coef_`.
3. **What attribute stores the y-intercept in Scikit-Learn Linear Regression?** `model.intercept_`.
4. **What triggers the goal state in the Water Jug problem?** When either jug volume matches the target requirement.
5. **Why is credit card fraud detection framed as a classification problem?** Because transactions are categorized into binary classes: Fraudulent (1) or Legitimate (0).

Slip 12

1. **What type of correlation exists between car age and selling price?** Negative correlation (price decreases as age increases).
2. **What function adds grid lines to a Matplotlib plot?** `plt.grid(True)`.
3. **What is the total move formula for n disks in Tower of Hanoi?** 2^n - 1 moves.
4. **What inputs comprise the perception space for a vacuum cleaner agent?** Current location coordinates and room cleanliness status.
5. **What basic actions can a vacuum cleaner agent perform?** `Suck` (Clean), `Move Left`, or `Move Right`.

Slip 17

1. **What function splits a dataset into training and testing sets in Scikit-Learn?** `train_test_split()`.
2. **What is the purpose of setting `random_state` in train-test splits?** To seed random shuffling so data splits remain reproducible.
3. **Why is a test set separated from the training set?** To evaluate the model's generalization performance on unseen data.
4. **How does tracking traffic density history help signal automation?** It enables adaptive timing adjustments based on traffic flow trends.
5. **What does Mean Squared Error (MSE) calculate in regression models?** The average squared difference between estimated and actual target values.

Slip 18

1. **What is a palindrome string?** A string that reads identically forwards and backwards.
2. **How does Python string slicing `string[::-1]` work?** It reverses the string by stepping backwards with a step size of `-1`.
3. **How does `.loc[]` assist in updating filtered DataFrame records?** It accesses specific rows and columns by label for targeted in-place assignments.

4. **What are the three core phases of an intelligent agent's execution cycle?** Perception \rightarrow Decision \rightarrow Action.
5. **What do actuators represent in intelligent agents?** The physical or virtual mechanisms used to execute agent decisions.

Slip 19

1. **How do you check the dimensions (rows and columns) of a DataFrame?** Using `df.shape`.
2. **How do you inspect the first N rows of a DataFrame?** Using `df.head(N)`.
3. **How do you check for missing values across a DataFrame?** Using `df.isnull().sum()`.
4. **What is class imbalance in classification datasets?** When one target class significantly outnumbers the other class samples.
5. **Why can class imbalance negatively affect machine learning models?** It can bias the model toward predicting the majority class.

Slip 20

1. **How does Multiple Linear Regression differ from Simple Linear Regression?** Multiple regression uses two or more independent predictor features instead of one.
2. **What independent features are used in the advertising sales dataset?** TV, Radio, and Newspaper advertising budgets.
3. **How do you check for duplicate rows in a Pandas DataFrame?** Using `df.duplicated()`.
4. **How do you remove duplicate rows from a DataFrame?** Using `df.drop_duplicates()`.
5. **What is vectorization in Pandas?** Performing operations on entire columns simultaneously without Python `for` loops.

Slip 21

1. **How does Logistic Regression convert continuous outputs into class labels?** By applying a probability threshold (typically 0.5).
2. **What does a prediction probability \ge 0.5 signify in binary classification?** It assigns the sample to class `1`.
3. **What detailed structural information does `df.info()` provide?** Column names, non-null counts, data types, index range, and memory usage.
4. **What defines a Goal-Based Agent?** An agent that selects actions designed to achieve explicit target goal states.

5. **How does a goal-based agent decide navigation steps along a path?** By comparing the index of its current location against the destination goal location index.